

An Exact Trace-Tail Formula for the De Blasi Measure in Finite von Neumann Preduals

Candidate solution packet for arXiv:1901.08056

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Status. Candidate full solution, likely valid, to the finite σ -finite clause of Question 12.1 in Hamhalter–Kalenda–Peralta–Pfitzner [1]. The general semifinite case is not claimed.

1 Source question

The source is J. Hamhalter, O. F. K. Kalenda, A. M. Peralta, and H. Pfitzner, *Measures of weak non-compactness in preduals of von Neumann algebras and JBW^* -triples*, arXiv:1901.08056. Question 12.1 appears on PDF page 54.

12. FINAL REMARKS AND OPEN PROBLEMS

Theorem 4.1 says that the three measures of weak non-compactness considered in this paper coincide in preduals of JBW^* -triples which complements the previous results from [46, 34]. However, the mentioned results include explicit formulas for these measures. In fact, these formulas are substantially used in the proofs. In the present paper we do not get an explicit formula due to the procedure of the proof – we use subsequence splitting property, Lemma 8.3, Proposition 3.2 and Lemma 2.1. So it is natural to ask whether there are some natural formulas for the De Blasi measure ω . The first question deals with a special case of von Neumann algebras.

Question 12.1. *Let M be a semifinite von Neumann algebra with a fixed normal semifinite faithful trace τ . Is there a formula for the De Blasi measure of weak noncompactness in M_* in terms of the trace τ ? Is it so at least for finite σ -finite von Neumann algebras?*

We note that the special case of commutative spaces $L^1(\mu)$ is settled in [46, Section 7].

The question asks for a formula, in terms of a normal semifinite faithful trace τ , for the De Blasi measure of weak noncompactness in M_* , and asks explicitly whether this can at least be done for finite σ -finite von Neumann algebras.

2 Definitions and statement

Let M be a finite σ -finite von Neumann algebra. Such an algebra admits a faithful normal finite trace; this is also recalled in Lemma 7.12(a) of the source paper [1]. Fix one such trace τ . We do not require $\tau(1) = 1$.

We use the standard isometric identification

$$M_* = L^1(M, \tau), \quad \phi_h(x) = \tau(hx), \quad \|\phi_h\| = \|h\|_1 = \tau(|h|).$$

For a bounded set $A \subset L^1(M, \tau)$, its De Blasi measure is

$$\omega(A) = \inf_K \sup_{h \in A} \text{dist}(h, K),$$

where K ranges over the relatively weakly compact subsets of $L^1(M, \tau)$.

For $\delta > 0$, define the right trace-tail quantity

$$u_{\tau, \delta}(A) = \sup_{h \in A} \sup \{ \|he\|_1 : e \in \text{Proj}(M), \tau(e) \leq \delta \},$$

and, by monotonicity,

$$u_\tau(A) = \lim_{\delta \downarrow 0} u_{\tau, \delta}(A) = \inf_{\delta > 0} u_{\tau, \delta}(A).$$

Theorem 1. *Let M be a finite σ -finite von Neumann algebra, let τ be a faithful normal finite trace on M , and let $A \subset M_* = L^1(M, \tau)$ be bounded. Then*

$$\boxed{\omega(A) = u_\tau(A)}.$$

Equivalently,

$$\omega(A) = \lim_{\delta \downarrow 0} \sup_{h \in A} \sup_{\substack{e \in \text{Proj}(M) \\ \tau(e) \leq \delta}} \|he\|_1.$$

Thus Question 12.1 has an affirmative answer for finite σ -finite von Neumann algebras.

The analogous left-tail formula follows either by the same proof or by applying the theorem to $A^* = \{h^* : h \in A\}$.

3 Proof Intuition

The proposed quantity vanishes precisely on weakly compact sets for a very concrete reason. The source paper proves that a weakly compact set in M_* is uniformly controlled by one normal state in the strong* topology. A small- τ projection, multiplied by any contraction, is uniformly small for that state; hence every functional in the weakly compact set has a small right tail on that projection. This immediately gives $u_\tau(A) \leq \omega(A)$.

For the reverse inequality, write $h = v|h|$ and cut $|h|$ off at height c . The discarded part is he , where $e = 1_{(c, \infty)}(|h|)$ and $\tau(e) \leq \|h\|_1/c$. The retained parts have uniformly bounded L^2 norm. Because the trace is finite, the inclusion $L^2(M, \tau) \rightarrow L^1(M, \tau)$ is bounded, so a bounded L^2 -ball becomes a weakly compact subset of L^1 . The trace tail is therefore exactly the distance scale needed in the De Blasi definition.

4 Proof

We first isolate the weak-compactness input.

Lemma 1. *If $K \subset L^1(M, \tau)$ is relatively weakly compact, then $u_\tau(K) = 0$.*

Proof. Replacing K by its weak closure, assume it is weakly compact. Theorem 11.1(b') of [1], specialized to the von Neumann algebra M , gives a normal state ψ such that the following holds: for every $\varepsilon > 0$ there is $\eta > 0$ such that

$$x \in B_M, \quad \|x\|_\psi < \eta \quad \implies \quad |\phi_h(x)| < \varepsilon \quad (h \in K), \quad (1)$$

where

$$\|x\|_\psi^2 = \psi(x^* \circ x) = \frac{1}{2} \psi(x^*x + xx^*).$$

Write $\psi(x) = \tau(gx)$ with $g \in L^1(M, \tau)_+$ and $\tau(g) = 1$.

We claim that

$$\sup_{\substack{e \in \text{Proj}(M), \tau(e) \leq \delta \\ a \in B_M}} \|ea\|_\psi \longrightarrow 0 \quad (\delta \downarrow 0). \quad (2)$$

Indeed, given $c > 0$, use functional calculus to write $g = g_0 + r$ with $0 \leq g_0 \leq c1$, $r \geq 0$, and $\|r\|_1 \rightarrow 0$ as $c \rightarrow \infty$. For $e \in \text{Proj}(M)$ and $a \in B_M$, traciality and positivity give

$$\begin{aligned} \psi(a^*ea) &= \tau(eaga^*) \leq c\tau(e) + \|r\|_1, \\ \psi(eaa^*e) &\leq \psi(e) \leq c\tau(e) + \|r\|_1. \end{aligned}$$

Consequently

$$\|ea\|_\psi^2 \leq c\tau(e) + \|r\|_1.$$

First choose c so that $\|r\|_1$ is small, and then choose δ so that $c\delta$ is small. This proves (2).

For such a small δ , apply (1) to $x = ea$. Since $\phi_h(ea) = \tau(hea)$ and $M = (L^1(M, \tau))^*$ isometrically,

$$\|he\|_1 = \sup_{a \in B_M} |\tau(hea)| < \varepsilon \quad (h \in K, e \in \text{Proj}(M), \tau(e) \leq \delta).$$

Thus $u_{\tau, \delta}(K) \rightarrow 0$. □

We now prove the two inequalities in Theorem 1.

Let K be relatively weakly compact and set

$$d = \sup_{h \in A} \text{dist}(h, K).$$

For $\zeta > 0$, each $h \in A$ has $k_h \in K$ with $\|h - k_h\|_1 < d + \zeta$. Right multiplication by a projection is contractive on L^1 , so

$$\|he\|_1 \leq \|(h - k_h)e\|_1 + \|k_he\|_1 \leq d + \zeta + u_{\tau, \delta}(K)$$

whenever $\tau(e) \leq \delta$. Lemma 1, followed by $\zeta \downarrow 0$, yields $u_\tau(A) \leq d$. Taking the infimum over K gives

$$u_\tau(A) \leq \omega(A). \quad (3)$$

For the reverse inequality, put $R = \sup_{h \in A} \|h\|_1$. The case $R = 0$ is trivial, so suppose $R > 0$. For $h = v|h|$ its polar decomposition and $c > 0$, set

$$e_{h,c} = 1_{(c, \infty)}(|h|), \quad k_{h,c} = v|h|1_{[0, c]}(|h|).$$

Then

$$\tau(e_{h,c}) \leq \frac{\|h\|_1}{c} \leq \frac{R}{c}, \quad (4)$$

and

$$\|h - k_{h,c}\|_1 = \|he_{h,c}\|_1 \leq u_{\tau,R/c}(A). \quad (5)$$

Moreover,

$$\|k_{h,c}\|_2^2 = \tau(|h|^2 1_{[0,c]}(|h|)) \leq c\tau(|h|) \leq cR. \quad (6)$$

Let $J : L^2(M, \tau) \rightarrow L^1(M, \tau)$ be the canonical inclusion. Since $\tau(1) < \infty$, noncommutative Cauchy–Schwarz gives $\|Jz\|_1 \leq \tau(1)^{1/2}\|z\|_2$, so J is bounded. The closed ball $\sqrt{cR} B_{L^2(M, \tau)}$ is weakly compact, and therefore its image

$$W_c = J(\sqrt{cR} B_{L^2(M, \tau)})$$

is weakly compact in $L^1(M, \tau)$. By (6), every $k_{h,c}$ lies in W_c . Equation (5) therefore implies

$$\omega(A) \leq \sup_{h \in A} \text{dist}(h, W_c) \leq u_{\tau,R/c}(A).$$

Letting $c \rightarrow \infty$ gives

$$\omega(A) \leq u_\tau(A). \quad (7)$$

Together, (3) and (7) prove the theorem.

5 Verification and edge cases

The script `code/verify_trace_tail.py` checks the spectral truncation identities (4)–(6), right-multiplication contractivity, and the finite-trace $L^2 \rightarrow L^1$ estimate on 240 random complex matrices. It is only a numerical sanity check; the proof above is analytic.

The finiteness of $\tau(1)$ is used exactly once: to make the inclusion $L^2(M, \tau) \rightarrow L^1(M, \tau)$ bounded. For a general semifinite trace, spectral truncation controls concentration on small-trace projections but not escape to regions of arbitrarily large trace. Thus the theorem does not claim the first, fully semifinite clause of Question 12.1; a separate trace-tightness term would be needed there.

6 Bounded novelty search

Before promotion, the run indexes were searched for arXiv:1901.08056 and the terms “De Blasi”, “weak non-compactness”, “von Neumann predual”, and “trace formula”. The only hit used arXiv:1901.08056 as background for a different operator-valued quantum-measure result. Web/arXiv searches on 21 July 2026 used the phrases “De Blasi measure noncommutative L1 projection trace formula”, “De Blasi von Neumann algebra L1 uniform integrability”, and “measure of weak noncompactness noncommutative L1 projection”. They found the commutative L^1 formula of Kačena–Kalenda–Spurný [2], the source paper, and later work on measures for operators, but no exact set formula of the form proved here for finite noncommutative $L^1(M, \tau)$. This is a bounded search, not a guarantee of novelty.

7 Human-review recommendation

The result should receive expert review. The two most important checks are: (i) the specialization of Theorem 11.1(b’) of [1] in Lemma 1, including the uniform estimate for ea ; and (ii) the use of the finite-trace L^2 ball as a weakly compact container for all spectral truncations. Subject to those checks, the proof gives a complete exact formula for the finite σ -finite case explicitly requested in Question 12.1.

References

- [1] J. Hamhalter, O. F. K. Kalenda, A. M. Peralta, and H. Pfitzner, *Measures of weak non-compactness in preduals of von Neumann algebras and JBW^* -triples*, arXiv:1901.08056 (2019).
- [2] M. Kačena, O. F. K. Kalenda, and J. Spurný, *Quantitative Dunford–Pettis property*, Adv. Math. 234 (2013), 488–527; arXiv:1110.1243.